

Sharp negative-direction summability for countable exponential feedback

A coefficient-to-remainder principle,
optimal logarithmic scales, and sectorial inversion

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Abstract

Consider the formal and sectorial equation

$$U(q) = \sum_{j \geq 1} q^j \exp(\lambda_j U(q)), \quad \lambda_j \geq 0.$$

A recent ProveIt draft establishes coefficient regularity and a left-sector realization but explicitly leaves uniform Gevrey remainders unproved. We close that gap for a broad class of weights. The change of unknown $U = qV$ gives a fixed dependent-variable disk about $V = 1$. An exact exponential-remainder estimate links the kernel to a positive subfamily of the Lagrange coefficients. We prove that the canonical sectorial solution has strong $\mathbb{M}$ -asymptotics if and only if its formal coefficients belong to $\mathcal{C}_{\mathbb{M}}$, under explicit elementary weight hypotheses. In particular, strong Gevrey- s asymptotics are equivalent to $\lambda_j = O((j \log(j+1))^{s+1})$ for every $s > 0$. The classical sectorial characterization of summability then gives a sharp application: for every $k > 1$, negative-direction k -summability holds if and only if $\lambda_j = O((j \log(j+1))^{1+1/k})$. For $\lambda_j \asymp j^p (\log(e+j))^\beta$, $p > 1$, we determine the optimal factorial-logarithmic weight, up to geometric factors, and prove actual truncation errors on the scale $\exp[-c|q|^{-1/(p-1)} (\log(1/|q|))^{(p-\beta)/(p-1)}]$. The results extend to the sectorial compositional inverse. Detailed proofs, error certificates, exact checks, limitations, and further research questions are included.

Status. These are conventional mathematical proofs, not Lean-verified theorems. The model-specific coefficient-to-remainder and summability conclusions are the proposed contribution. General weighted implicit-function methods and the classical summability theorem are not claimed as inventions. Global novelty and priority have not been certified. No named longstanding conjecture is claimed solved.

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1 The research gap and the main conclusions

1.1 A precise continuation of the repository

The ProveIt snapshot used here is commit

0c973d8f5e7ef4550f4df1bf486d890037fd9f14.

Its transseries documentation is in `Analysis/Transseries/docs/series-and-transseries/`. The directory README distinguishes the consolidated volumes from nineteen unmerged research arrivals dated 29 September 2026, and warns that those arrivals have not been reviewed claim by claim or formalized [1].

The immediate predecessor is *A sharp regularity classification for countable exponential-feedback transseries*, in `Exponential_Feedback_Regularity_Classification/` [2]. It studies

$$\widehat{U}(q) = \sum_{j \geq 1} q^j \exp(\lambda_j \widehat{U}(q)), \quad \widehat{U}(q) = \sum_{n \geq 1} u_n q^n, \quad \lambda_j \geq 0. \quad (1.1)$$

Its README and sectorial-realization proof explicitly distinguish Poincaré asymptotics from uniform Gevrey remainder estimates. For each fixed N , the assertion $R_N(q) = O_N(q^N)$ alone says nothing useful about how the constant grows with N .

The neighboring *Finite-Core Universality and Sharp Large Order* draft develops multiplicative coefficient asymptotics and finite-core condensation [3]. We do not repeat its saddle analysis. The present question is to control the actual error of a sectorial solution and identify which summation procedures reconstruct it.

Editorial note (ProveIt, 2026-09-29). The snapshot used here already contained a third package, `Negative_Ray_Summation_Exponential_Feedback/` [8], which the directory README described and which this article does not cite. It proves, for $s = 1$ and unit amplitudes, that $\widehat{U}$ and its compositional inverse are *finely* (fixed-width half-strip) Borel–Laplace summable in direction π exactly when $\lambda_j = O((j \log j)^2)$ (its Theorem `thm:threshold`), including the quadratic model, by inverting the kernel before taking Borel transforms; it also continues the inverse Borel transform explicitly and, for $\lambda_j = j^2$, bounds the inverse remainder uniformly on a closed left half-disc (its Theorems `thm:borel` and `thm:remainder`). The two articles are complementary: this one covers every $s > 0$ and sectorial k -summability for $k > 1$; that one covers the fine case $k = 1$; both prove the Gevrey-1 class equivalence. The ordinary (angular) case $k = 1$ for $\lambda_j = j^2$ is settled negatively by the later package `Natural_Boundaries_Quadratic_Exponential_Feedback/` [9]. All three packages are in `Analysis/Transseries/docs/series-and-transseries/`.

1.2 Principal statements

Theorem 1.1 (Sharp sectorial Gevrey and summability classification). *Let (λ_j) be any sequence of finite nonnegative real numbers. There is a canonical holomorphic germ U on a left sectorial region of opening π , characterized by (1.1) and $U = q + O(q^2)$. For every $s > 0$, the following conditions are equivalent:*

- (i) $\lambda_j = O((j \log(j+1))^{s+1})$;
- (ii) $|u_n| \leq C A^n (n!)^s$ for some $C, A > 0$;

(iii) on every proper left sector, after choosing a sufficiently small radius, there are $C, A > 0$ such that for all $N \geq 1$,

$$\left| U(q) - \sum_{n=1}^{N-1} u_n q^n \right| \leq C A^N (N!)^s |q|^N. \quad (1.2)$$

For every $k > 1$, these conditions with $s = 1/k$ are also equivalent to $\widehat{U}$ being k -summable in direction π . Its sum is U . The same equivalences hold for the formal and sectorial compositional inverses of U .

The summability conclusion deliberately assumes $k > 1$; the case $k = 1$ is not settled here. The remainder assertion holds for every $s > 0$.

Editorial note (ProveIt, 2026-09-29). Scope: the word “Sharp” in the title refers to these equivalences. For summability they hold for $k > 1$ only, that is for subquadratic slope classes; the quadratic model, which is the regularity article’s directional question, is not decided here. Its fine $k = 1$ summability is proved in `Negative_Ray_Summation_Exponential_Feedback/ [8]`, and its angular 1-summability fails for $\lambda_j = j^2$ by `Natural_Boundaries_Quadratic_Exponential_Feedback/ [9]`.

Theorem 1.2 (Optimal logarithmic scale). *Suppose, for some $p > 1$, $\beta \in \mathbb{R}$, and $c, C > 0$, that*

$$c j^p (\log(e + j))^\beta \leq \lambda_j \leq C j^p (\log(e + j))^\beta \quad \text{for all sufficiently large } j. \quad (1.3)$$

Set $s = p - 1$, $\tau = \beta - p$, and, up to a finite initial quotient adjustment, define

$$M_n = (n!)^s \prod_{\ell=1}^n (\log(e + \ell))^\tau. \quad (1.4)$$

Then U and its sectorial inverse have strong $\mathbb{M}$ -asymptotic expansions. This weight cannot be replaced by $M_{s, \tau - \varepsilon}$, $\varepsilon > 0$, or by $M_{s', \tau'}$ with $0 < s' < s$ and fixed τ' . On every proper left sector there is a truncation order $N = N(|q|)$ giving

$$\left| U(q) - \sum_{n < N} u_n q^n \right| \leq C_1 \exp \left[-c_1 |q|^{-1/s} (\log(1/|q|))^{-\tau/s} \right]. \quad (1.5)$$

The inverse has the same error scale, with possibly different constants. If $1 < p < 2$, both series are also $\mathbb{M}$ -summable in direction π in the standard generalized sense.

Here optimality is within the factorial–logarithmic weight scale, up to geometric factors. It is not a lower bound on each pointwise remainder, and does not specify the sharp numerical constant c_1 .

1.3 Why a change of unknown is necessary

If λ_j/j is unbounded, the original kernel is not analytic in a product neighborhood of $(q, U) = (0, 0)$. At fixed positive U , its terms fail to tend to zero along a subsequence for every positive q . An implicit-function theorem applied directly in these coordinates would therefore lack a required hypothesis. Instead put

$$U = qV, \quad F(q, v) = \sum_{j \geq 1} q^{j-1} e^{\lambda_j q v}, \quad V = F(q, V). \quad (1.6)$$

On a proper left sector, qv remains in the left half-plane throughout one fixed disk about $v = 1$. All exponentials are bounded by one, independently of the slope sizes. This fixed dependent-variable disk is the essential missing hypothesis, not a new formal inversion formula.

Gevrey asymptotic implicit-function theory is classical; for example Nikolaev establishes a Gevrey implicit-function and resummation theorem under appropriate uniform assumptions [4]. We verify a weighted version directly for this countable singular kernel and use positivity to obtain an equivalence rather than only sufficiency.

2 Conventions and admissible weights

For $0 < \eta < 1$ and $r > 0$, write

$$S_{\eta,r} = \{q \in \mathbb{C} : 0 < |q| < r, \operatorname{Re} q < -\eta|q|\}. \quad (2.1)$$

Its opening is $2\arccos \eta < \pi$. A left sectorial germ means compatible functions on these sectors, with radii allowed to depend on η . No radius uniform as $\eta \downarrow 0$ is asserted.

Definition 2.1 (Admissible weights). A sequence $\mathbb{M} = (M_n)_{n \geq 0}$ is admissible here if $M_0 = 1$, its quotients $\mu_n = M_n/M_{n-1}$ satisfy

$$1 \leq \mu_1 \leq \mu_2 \leq \cdots, \quad \mu_n \longrightarrow \infty, \quad (2.2)$$

and, for some $D, E \geq 1$,

$$\mu_n^n \leq D^n M_n \quad (n \geq 1), \quad M_{n+1} \leq E^{n+1} M_n \quad (n \geq 0). \quad (2.3)$$

These are the precise hypotheses used in our proof, not a proposed replacement for the standard definition of strong regularity. Familiar factorial–logarithmic weights satisfy them; standard strongly regular weights provide many examples [5].

Definition 2.2 (Coefficient and strong asymptotic classes). A formal series $\hat{f} = \sum f_n q^n$ belongs to $\mathcal{C}_{\mathbb{M}}$ if $|f_n| \leq CA^n M_n$ for some constants. A function f has a strong $\mathbb{M}$ -asymptotic expansion $\hat{f}$ on $S_{\eta,r}$ if

$$\left| f(q) - \sum_{n=0}^{N-1} f_n q^n \right| \leq CA^N M_N |q|^N \quad (N \geq 1, q \in S_{\eta,r}), \quad (2.4)$$

with constants independent of N and q . For a sectorial germ this is required on every proper left sector, allowing the radius and constants to depend on that sector.

Lemma 2.3 (Elementary weight facts). *For an admissible weight,*

$$M_a M_b \leq M_{a+b}, \quad M_n \leq \mu_N^n \quad (0 \leq n \leq N). \quad (2.5)$$

Multiplication by q and, for zero-constant-term expansions, division by q preserve the coefficient and strong asymptotic classes. Strong $\mathbb{M}$ -asymptotics imply formal membership in $\mathcal{C}_{\mathbb{M}}$.

Proof. Multiply the ordered quotients to obtain the inequalities. Multiplication by q shifts an index down, and $M_{n-1} \leq M_n$; division shifts up, which is absorbed by the bound involving E . For coefficient necessity, divide the remainder after degree $N - 1$ by q^N and approach zero along an interior ray. The expansion one order further identifies the limit as f_N , and gives $|f_N| \leq CA^N M_N$. $\square$

Lemma 2.4 (Factorial–logarithmic weights). *For $s > 0$ and $\tau \in \mathbb{R}$, the weight (1.4), with a finite initial quotient adjustment if needed, is admissible, and*

$$\mu_n \sim n^s (\log n)^\tau, \quad (2.6)$$

$$\log M_n = sn \log n - sn + \tau n \log \log n + o(n). \quad (2.7)$$

It is geometrically equivalent to $(n!)^s (\log(e+n))^{\tau n}$.

Proof. The function $x^s (\log(e+x))^\tau$ is increasing and at least one for all sufficiently large x . Keep these quotients from an index J onward and choose nondecreasing quotients between 1 and μ_J before that index. This changes the later weights by a constant factor. Stirling's formula and

$$\sum_{\ell=1}^n \log \log(e+\ell) = n \log \log n + O(n/\log n)$$

give (2.7). The sum estimate follows by splitting at $\sqrt{n}$ and comparing with an integral, then integrating by parts. Thus $n \log \mu_n - \log M_n = sn + o(n)$, which proves the D bound. The quotient μ_{n+1} has polynomial times logarithmic growth and is bounded by E^{n+1} after enlarging E . The same sum estimate proves geometric equivalence with the simpler weight. $\square$

3 A weighted implicit remainder theorem

The uniform-in-order step uses only Cauchy estimates, contraction, and an analytic majorant. No summability hypothesis on the kernel is used.

Theorem 3.1 (Uniform weighted implicit expansion). *Let $\mathbb{M}$ be admissible. Suppose F is holomorphic on $S_{\eta, r_0} \times D(1, \delta)$, with coefficient functions a_n holomorphic in the disk, $a_0 = 1$, and uniform bounds*

$$|a_n(v)| \leq CA^n M_n \quad (n \geq 1), \quad (3.1)$$

$$\left| F(q, v) - \sum_{n=0}^{N-1} a_n(v) q^n \right| \leq CA^N M_N |q|^N \quad (N \geq 1). \quad (3.2)$$

After shrinking the radius, the fixed point $V = F(q, V)$ near 1 exists, is holomorphic, and has a strong $\mathbb{M}$ -expansion equal to the unique formal solution. Constants can be obtained from the stated data.

Proof. Increase constants so that $A, C, D \geq 1$.

Fixed-disk existence. The $N = 1$ estimate gives $F(q, v) = 1 + O(q)$ uniformly. Cauchy's estimate gives $\partial_v F = O(q)$ on $D(1, \delta/2)$. Shrinking the radius makes F preserve the closed smaller disk with Lipschitz constant at most $1/2$. Iteration gives a unique fixed point, holomorphic by locally uniform convergence.

Formal majorization. Write $V = 1 + w$. Cauchy's estimate gives

$$a_n(1+w) = \sum_{j \geq 0} a_{n,j} w^j, \quad |a_{n,j}| \leq CA^n M_n \delta^{-j}.$$

In total degree N , the weights of a term $q^n w_{i_1} \cdots w_{i_j}$ are bounded by $M_n M_{i_1} \cdots M_{i_j} \leq M_N$. Induction therefore dominates $|w_N|/M_N$ by $[z^N]H(z)$, where

$$H = \frac{CAz}{1-Az} \frac{1}{1-H/\delta}, \quad H(0) = 0. \quad (3.3)$$

This analytic scalar equation is contractive on $|H| \leq \delta/2$ for $|z| \leq \min\{(2A)^{-1}, \delta/(16CA)\}$. Its coefficients are nonnegative. Thus there are $K, B \geq 1$ such that the unique formal solution $\hat{V} = \sum v_n q^n$ satisfies

$$|v_n| \leq KB^n M_n \quad (n \geq 0). \quad (3.4)$$

Formal uniqueness follows because every nonconstant kernel term has positive q -order.

Shrinking analytic disks. Put $F_{<N} = \sum_{n < N} a_n(v)q^n$ and $\rho_N = c/(A\mu_N)$, where $0 < c \leq \min\{1/4, \delta/(16C)\}$. On $|q| \leq \rho_N$,

$$\sum_{n=1}^{N-1} CA^n M_n |q|^n \leq C \sum_{n \geq 1} c^n \leq Cc/(1-c).$$

Cauchy's estimate in v makes $F_{<N}$ contractive on $\overline{D}(1, \delta/2)$, decreasing c if needed. Its fixed point V_N is analytic in $D(0, \rho_N)$ and has the same Taylor coefficients as $\hat{V}$ through degree $N-1$. For sectorial q with $|q| \leq \rho_N/2$, subtract the fixed-point equations to get

$$|V - V_N| \leq 2CA^N M_N |q|^N. \quad (3.5)$$

Cauchy's estimate for $V_N - 1$, bounded by $\delta/2$, gives

$$\left| V_N - \sum_{n < N} v_n q^n \right| \leq \delta(|q|/\rho_N)^N \leq \delta(AD/c)^N M_N |q|^N. \quad (3.6)$$

This proves the estimate in the small- q range associated with N .

The complementary range. When $|q| \geq \rho_N/2$, for $0 \leq n < N$,

$$\frac{M_n |q|^n}{M_N |q|^N} = \frac{M_n}{\mu_N^n} \frac{\mu_N^N}{M_N} (|q|\mu_N)^{-(N-n)} \leq D^N (2A/c)^{N-n}. \quad (3.7)$$

Combine this with (3.4) and the boundedness of V . Summing and using $N+1 \leq 2^N$ bounds the normalized remainder by a constant times $[2D \max\{B, 2A/c, 1\}]^N$. Together with (3.5)–(3.6), this is a single all-orders strong remainder estimate on the smaller sector. $\square$

Remark 3.2. This proof permits a geometric enlargement of the weight. It preserves class membership, not necessarily the optimal numerical coefficient or remainder type. The shrinking-disk Cauchy estimate is one explicit source of such a loss of constants.

4 The desingularized countable kernel

Fix $0 < \eta < 1$, choose $0 < \delta < \eta$, and put $c = \eta - \delta > 0$, $B = 1 + \delta$. For $q \in S_{\eta,r}$ and $|v - 1| < \delta$,

$$\operatorname{Re}(qv) \leq -c|q|. \quad (4.1)$$

All the following estimates are uniform on this product domain.

Proposition 4.1 (Sectorial existence with arbitrary slopes). *For every nonnegative slope sequence, F in (1.6) is holomorphic on $S_{\eta,r} \times D(1, \delta)$ when $r < 1$, and*

$$|F(q, v) - 1| \leq \lambda_1 B |q| + \frac{|q|}{1 - |q|}, \quad (4.2)$$

$$|\partial_v F(q, v)| \leq \lambda_1 |q| + \frac{|q|}{ec(1 - |q|)}. \quad (4.3)$$

For a sufficiently small radius it has a unique fixed point near 1. The corresponding $U = qV$ solves (1.1), has $U = q + O(q^2)$, and is unique among solutions with that behavior on a proper left sector. Constructions on different sectors agree near zero on their overlaps.

Proof. The j th term is bounded by $|q|^{j-1}$, giving normal convergence. For $j = 1$ use $e^z - 1 = z \int_0^1 e^{tz} dt$ and $\operatorname{Re} z \leq 0$; the remaining terms have a geometric majorant. For the derivative handle $j = 1$ directly. For $j \geq 2$, the inequality $te^{-ct|q|} \leq 1/(ec|q|)$, $t \geq 0$, bounds the derivative term by $|q|^{j-1}/(ec)$. Summing proves the two estimates. They tend to zero, giving contraction on a fixed closed disk about 1. The first estimate then gives $V = 1 + O(q)$. Any other solution with the stated behavior has U/q in the same disk near zero, proving uniqueness and compatibility. $\square$

Proposition 4.2 (Exact uniform kernel remainder). *Define*

$$a_n(v) = \sum_{j=1}^{n+1} \frac{(\lambda_j v)^{n+1-j}}{(n+1-j)!}, \quad n \geq 0. \quad (4.4)$$

Then $a_0 = 1$, $|a_n(v)| \leq a_n(B)$, and for every $N \geq 1$,

$$\left| F(q, v) - \sum_{n=0}^{N-1} a_n(v) q^n \right| \leq \frac{a_N(B)}{1 - |q|} |q|^N. \quad (4.5)$$

No growth restriction on the slopes is needed.

Proof. For $\operatorname{Re} z \leq 0$, $m \geq 1$, the integral Taylor remainder gives

$$\left| e^z - \sum_{h=0}^{m-1} \frac{z^h}{h!} \right| = \left| \frac{z^m}{(m-1)!} \int_0^1 (1-t)^{m-1} e^{tz} dt \right| \leq \frac{|z|^m}{m!}. \quad (4.6)$$

For $j \leq N$, apply this with $z = \lambda_j qv$ and $m = N + 1 - j$, then multiply by q^{j-1} . The combined bound is

$$|q|^N \sum_{j=1}^N \frac{(B\lambda_j)^{N+1-j}}{(N+1-j)!}.$$

The terms $j > N$ contribute at most $|q|^N/(1 - |q|)$. The missing $j = N + 1$ summand of $a_N(B)$ equals 1, so their sum is bounded by (4.5). $\square$

The estimate holds on a fixed disk of dependent-variable values, not only along the unknown solution. This is the uniformity required in Theorem 3.1. Substitution of a formal polynomial into the original kernel would not establish it.

5 Positivity and the coefficient-to-remainder principle

The following algebraic comparison does not depend on the sharper coefficient asymptotics in the predecessor drafts.

Lemma 5.1 (Formal coefficients and a positive subfamily). *Equation (1.1) has a unique solution in $q\mathbb{R}[[q]]$. It has $u_1 = 1$, $u_n \geq 1$, and*

$$u_n = \sum_{\substack{\mathbf{m} \text{ finite} \\ \sum_j m_j = n}} \frac{(\sum_j \lambda_j m_j)^{|\mathbf{m}|-1}}{\prod_j m_j!}, \quad |\mathbf{m}| = \sum_j m_j, \quad (5.1)$$

where $0^0 = 1$. In particular,

$$a_n(1) \leq u_{n+1} \quad (n \geq 0), \quad (5.2)$$

and for $j \geq 2$, $h \geq 0$,

$$u_{j+h} \geq \frac{\lambda_j^h}{h!}. \quad (5.3)$$

Proof. The defining map raises the order of a difference of formal candidate solutions by at least one, giving existence and uniqueness by iteration. Introduce a parameter t and solve $U = t\Phi(q, U)$, where $\Phi(q, z) = \sum_{j \geq 1} q^j e^{\lambda_j z}$. Lagrange inversion gives

$$U = \sum_{k \geq 1} \frac{t^k}{k} [z^{k-1}] \Phi(q, z)^k.$$

At each q -degree all sums are finite. Expanding by multiplicities cancels the multinomial and derivative factorials, yielding (5.1) when $t = 1$.

All terms are nonnegative. One action j and h actions 1 contribute $(\lambda_j + h\lambda_1)^h/h!$, proving (5.3). For (5.2), put $N = n + 1$. The configurations with one action $j \geq 2$ and $N - j$ actions 1 are distinct as j varies. They dominate the corresponding terms of $a_{N-1}(1)$. The all-action-1 configuration contributes $(N\lambda_1)^{N-1}/N!$, at least $\lambda_1^{N-1}/(N-1)!$ for $N \geq 2$, since their ratio when nonzero is N^{N-2} . The case $N = 1$ is equality. The single action n also proves $u_n \geq 1$. $\square$

Theorem 5.2 (Coefficient-to-remainder equivalence). *Let $\mathbb{M}$ be admissible and U the canonical left-sector solution. The following statements are equivalent:*

- (i) $\widehat{U} \in \mathcal{C}_{\mathbb{M}}$;
- (ii) $a_n(1) \leq CA^n M_n$ for some $C, A > 0$;
- (iii) U has strong $\mathbb{M}$ -asymptotics $\widehat{U}$ on every proper left sector;
- (iv) U has that strong expansion on at least one proper left sector.

Proof. By (5.2) and the shift bound for M_{n+1} , (i) implies (ii). Since $a_n(B) \leq B^n a_n(1)$ for $B \geq 1$, Proposition 4.2 and Theorem 3.1 turn (ii) into strong $\mathbb{M}$ -asymptotics of $V = U/q$ on every proper left sector. Multiplication by q preserves the class, and formal uniqueness identifies the coefficients; hence (iii). The implication (iii) to (iv) is immediate, and (iv) to (i) is coefficient necessity from Lemma 2.3. $\square$

The theorem proves that the canonical realization introduces no loss of weight class beyond geometric constants. Conversely, failure of formal membership rules out the corresponding strong expansion on every proper left sector. Positivity is essential in the present proof: it prevents cancellation from hiding a large kernel coefficient.

6 Determining the factorial–logarithmic scale

We prove the class estimates independently, without invoking an exact numerical Gevrey type or a multiplicative saddle equivalent from the repo.

Lemma 6.1 (Growth of the kernel coefficients). *Fix $p > 1$, $\beta \in \mathbb{R}$. If $\lambda_j \leq Cj^p(\log(e + j))^\beta$ eventually, then for each fixed $B \geq 1$ there are $K, A > 0$ such that*

$$a_n(B) \leq KA^n M_{p-1, \beta-p; n}. \quad (6.1)$$

Under the two-sided estimate (1.3),

$$\log a_n(1) = (p-1)n \log n + (\beta-p)n \log \log n + O(n). \quad (6.2)$$

Proof. Enlarge a constant to cover all exceptional slopes. Put $L = \log n$ and consider the term with index j and exponent $h = n + 1 - j$. The term $h = 0$ is 1. For $h \geq 1$, using $\log h! \geq h \log h - h$ gives

$$\log \frac{(B\lambda_j)^h}{h!} \leq h\{C_0 + p \log j + \beta \log \log(e + j) - \log h\}. \quad (6.3)$$

We bound the maximum uniformly in j .

For $j < \sqrt{n}$, the right side is at most $(p/2 - 1)nL + O(n \log L)$, smaller than the claimed bound by a positive multiple of nL . For $j > n/2$, $h \leq n/2 + 1$ and $\log \lambda_j \leq pL + O(\log L)$. The function $h(pL + O(\log L) - \log h + 1)$ increases on this range for large n , so its maximum is at most $\frac{1}{2}(p - 1)nL + O(n \log L)$, also smaller than required.

In the middle range $\sqrt{n} \leq j \leq n/2$, $\log \log(e + j) = \log L + O(1)$ and $\log h = L + O(1)$. Expanding $h = n + 1 - j$, with $s = p - 1$, gives

$$\begin{aligned} \log \frac{(B\lambda_j)^h}{h!} &\leq snL + \beta n \log L + pn \log(j/n) - sjL \\ &\quad + |\beta|j \log L - pj \log(j/n) + O(n) \\ &\leq snL + \beta n \log L + pn \log(j/n) - \frac{s}{2}jL + O(n). \end{aligned}$$

We used $-j \log(j/n) \leq n/e$ and $|\beta| \log L \leq sL/2$ eventually. Setting $t = jL/n$ turns the last line into

$$snL + (\beta - p)n \log L + n\{p \log t - st/2\} + O(n).$$

The expression in braces is bounded above for $t > 0$. Summing $n + 1$ terms changes the logarithm by at most $\log(n + 1)$, and Lemma 2.4 proves the upper bound.

For a lower bound, take $j = \lfloor n/\log n \rfloor$ and $h = n + 1 - j$. The two-sided slope assumption gives

$$\log \lambda_j = p \log n + (\beta - p) \log \log n + O(1), \quad h = n + O(n/\log n).$$

Stirling's formula for $h!$ yields the matching lower bound in (6.2). Zero slopes in the upper-bound argument are simply omitted. $\square$

Theorem 6.2 (Exact Gevrey class criterion). *For every $s > 0$,*

$$\widehat{U} \in \mathcal{C}_{(n!)^s} \iff \lambda_j = O((j \log(j + 1))^{s+1}). \quad (6.4)$$

Both conditions are equivalent to strong Gevrey- s asymptotics of the canonical left-sector solution.

Proof. For sufficiency use Lemma 6.1 with $p = s + 1$, $\beta = p$, giving logarithmic exponent zero, and then Theorem 5.2. For necessity suppose $u_n \leq CA^n(n!)^s$. For each large $j \geq 2$ set $h = \lceil j \log j \rceil$ and $n = j + h$. By (5.3),

$$\lambda_j^h \leq h!CA^n(n!)^s, \quad \lambda_j \leq C^{1/h}A^{n/h}h n^{sn/h}.$$

Now $n/h = 1 + O(1/\log j)$, $n \asymp j \log j$, and $n^{s(n/h-1)} = O(1)$. Thus $\lambda_j \leq C'(j \log j)^{s+1}$. $\square$

Proposition 6.3 (The analytic endpoint). *The formal solution has positive convergence radius exactly when $\lambda_j = O(j)$.*

Proof. If $\lambda_j \leq Cj$, the original kernel converges normally in a small bidisk, its terms bounded by $(|q|e^{C|U|})^j$. The analytic implicit-function theorem applies at $(0, 0)$. Conversely an exponential bound $u_n \leq CA^n$ and (5.3) with $h = j$ give $\lambda_j \leq C^{1/j}A^2(j!)^{1/j} = O(j)$. $\square$

The positive- s criterion must not be extrapolated to $s = 0$. For $\lambda_j = j \log(j+1)$ the series is divergent but belongs to every positive Gevrey class, and its canonical realization has each corresponding strong expansion.

Theorem 6.4 (Optimality within the logarithmic scale). *Under (1.3), with $s = p-1$, $\tau = \beta-p$,*

$$\log u_n = sn \log n + \tau n \log \log n + O(n). \quad (6.5)$$

The canonical solution has strong $\mathbb{M}_{s,\tau}$ -asymptotics, but no strong $\mathbb{M}_{s,\tau-\varepsilon}$ -asymptotics for $\varepsilon > 0$ on any proper left sector. Decreasing s cannot be compensated by any fixed logarithmic exponent.

Proof. The upper bound follows from Lemma 6.1 and the formal majorant in Theorem 3.1; the lower bound follows from $a_{n-1}(1) \leq u_n$ and (6.2). Shifting the index by one changes the expression by $o(n)$. Strong membership follows from Theorem 5.2. Reducing τ by ε subtracts $\varepsilon n \log \log n$ from the logarithm of the weight, which no geometric factor can absorb. Reducing s creates a larger deficit of order $n \log n$. Coefficient necessity rules out both improvements. $\square$

7 Actual optimal-truncation bounds

Define the associated function

$$h_{\mathbb{M}}(t) = \inf_{N \geq 0} M_N t^N, \quad t > 0. \quad (7.1)$$

The all-orders estimates now allow the truncation index to vary with q .

Proposition 7.1 (Associated error scale). *For $\mathbb{M}_{s,\tau}$ let $n(t)$ be an integer minimizing $M_N t^N$. As $t \downarrow 0$,*

$$n(t) \sim s^{\tau/s} t^{-1/s} (\log(1/t))^{-\tau/s}, \quad (7.2)$$

$$\log h_{\mathbb{M}}(t) \sim -sn(t). \quad (7.3)$$

Proof. Consecutive terms have ratio $t\mu_{N+1}$; a minimum occurs at the crossing of one. Since adjacent quotient ratios tend to one, $tn(t)^s (\log n(t))^\tau \rightarrow 1$. Substituting the right side of (7.2) verifies this inverse asymptotic. Its logarithm has leading term $\log(1/t)/s$. At that index, (2.7) gives

$$\log(M_{n(t)} t^{n(t)}) = n(t) \{s \log n(t) + \tau \log \log n(t) + \log t - s\} + o(n(t)) = -sn(t) + o(n(t)).$$

A one-index tie does not affect the equivalent. $\square$

Corollary 7.2 (Realized small errors). *If U has a strong $\mathbb{M}_{s,\tau}$ bound with geometric constant A , choose $N = n(A|q|)$. Then*

$$\left| U(q) - \sum_{n < N} u_n q^n \right| \leq C \exp[-c|q|^{-1/s} (\log(1/|q|))^{-\tau/s}] \quad (7.4)$$

for sufficiently small q on the chosen proper sector.

Proof. Take the infimum of the strong all-orders estimate and apply Proposition 7.1. The geometric constant changes the positive constant in the exponential, not its power or logarithmic correction. $\square$

For quadratic slopes $\lambda_j = j^2$, $s = 1$, $\tau = -2$, so

$$\left| U(q) - \sum_{n < N(q)} u_n q^n \right| \leq C \exp \left[-c \frac{(\log(1/|q|))^2}{|q|} \right], \quad N(q) \asymp \frac{(\log(1/|q|))^2}{|q|}. \quad (7.5)$$

This is an actual error estimate, not merely a formal least-term law. It implies strong Gevrey-1 bounds with arbitrarily small positive geometric type, since $A^N (\log(N+e))^{-2N} \leq C_\varepsilon \varepsilon^N$ for every $\varepsilon > 0$. The Taylor series is nevertheless divergent.

Under $q = e^{-x}$ the formal solution is $\sum_{n \geq 1} u_n e^{-nx}$. Proper left sectors correspond to horizontal strips about odd multiples of π in $\Im x$. With $X = \operatorname{Re} x$, (7.4) becomes

$$O(\exp[-ce^{X/s} X^{-\tau/s}]), \quad N(X) \asymp e^{X/s} X^{-\tau/s}. \quad (7.6)$$

These are complex-strip realizations, not claims on the real- x axis.

8 Sharp negative-direction summability

We use one classical external analytic characterization, with its hypotheses verified by the preceding proofs.

Classical sectorial characterization. For $s = 1/k > 0$, a formal series is k -summable in direction d exactly when it is the strong Gevrey- s expansion of a holomorphic function on a sectorial region centered at d with opening strictly greater than πs . The function is the unique sum there. Equivalently its moment-normalized Borel transform

$$\mathcal{B}_k \hat{f}(\xi) = \sum_{n \geq 0} \frac{f_n \xi^n}{\Gamma(1 + n/k)} \quad (8.1)$$

has the required analytic continuation and exponential-order- k growth near direction d , and generalized Laplace reconstruction recovers the function. We use the local-sector formulation in Lastra–Malek–Sanz, Definition 3.3, Theorem 3.18, and Remark 3.19(i) [5]. The weights $(n!)^{1/k}$ and $\Gamma(1 + n/k)$ are geometrically equivalent.

Theorem 8.1 (Exact summability criterion for $k > 1$). *For every $k > 1$,*

$$\boxed{\hat{U} \text{ is } k\text{-summable in direction } \pi \iff \lambda_j = O((j \log(j+1))^{1+1/k}).} \quad (8.2)$$

Whenever these conditions hold, its sum is the canonical solution U .

Proof. Set $s = 1/k < 1$. The slope condition and Theorem 6.2 give strong Gevrey- s asymptotics on every proper left sector. Choose η so small that $2 \arccos \eta > \pi s$. The classical characterization applies there and identifies the sum with U . Conversely k -summability implies a Gevrey-1/ k coefficient bound. Necessity in Theorem 6.2 gives the slope condition. $\square$

Corollary 8.2 (A logarithmically refined sum). *Under (1.3) with $1 < p < 2$, the formal solution is $\mathbb{M}_{p-1, \beta-p}$ -summable in direction π in the standard generalized ultraholomorphic sense. Its generalized sum is U .*

Proof. The factorial–logarithmic weights, with equivalent initial normalization, are strongly regular and have opening index $s = p-1$; these weight facts and their generalized reconstruction theory are established in [5, 6]. Theorem 6.4 supplies strong asymptotics for this weight. Since $s < 1$, an available left sector has opening greater than πs , so the characterization applies. $\square$

For $\lambda_j = j^p$, $1 < p < 2$, the critical ordinary summation order is $k = 1/(p - 1)$. For slopes comparable to $j^p(\log(e + j))^\beta$, ordinary summability at that critical order holds exactly when $\beta \leq p$. When $\beta > p$ it fails, but the logarithmically refined sum still exists. Every ordinary order satisfying $p - 1 < 1/k < 1$ remains available.

8.1 The critical aperture is not resolved by this argument

For $s = 1$ the directly available sectorial region has opening π , not strictly greater. Strong Gevrey-1 estimates on its proper subsectors therefore do not establish ordinary Borel summability by this theorem. The quadratic type-zero improvement does not give uniform control up to the imaginary boundary rays.

A Nevanlinna–Sokal criterion on a suitable tangent-disk domain could be relevant, but its geometric and uniformity hypotheses have not been established here [7]. Thus ordinary negative-direction Borel summability for $\lambda_j = j^2$ remains a stated research question, not an implicit corollary.

Editorial note (ProveIt, 2026-09-29). This question is settled for $\lambda_j = j^2$ by two other packages of the series. In the fine (half-strip) sense the answer is positive: `Negative_Ray_Summation_Exponential_Feedback/ [8]`, Theorem `thm:threshold`, available at this article’s snapshot. In the ordinary (angular) sense it is negative: `Natural_Boundaries_Quadratic_Exponential_Feedback/ [9]`, Theorem `thm:main`, proves that the literal inverse has a natural boundary on the imaginary diameter and that neither $\widehat{U}$ nor its inverse is 1-summable in any open arc of directions containing π . Uniform estimates for U up to the boundary rays are not provided by either.

8.2 The literal positive-real obstruction

Proposition 8.3. *If $\sup_j \lambda_j/j = \infty$, the literal infinite sum (1.1) has no solution for all sufficiently small positive real q with $U(q)/q \rightarrow 1$.*

Proof. Such a solution would have $\operatorname{Re} U(q) > 0$ for sufficiently small positive q . Fix one such q and choose a subsequence with $\lambda_j/j \rightarrow \infty$. Then

$$\log |q^j e^{\lambda_j U(q)}| = j \log q + \lambda_j \operatorname{Re} U(q) \longrightarrow +\infty$$

along that subsequence, so the summands do not tend to zero. □

This concerns convergence of the defining kernel. Analytic continuation of a function constructed elsewhere, or a generalized sum in another direction, is a different question. We do not infer failure of every possible continuation from this test.

9 Sectorial compositional inversion

The inverse result gives an actual inverse germ with strong remainders, not merely a transfer of a formal regularity label.

Lemma 9.1 (Formal weighted reversion). *For an admissible weight and $\widehat{f}(q) = q + \sum_{n \geq 2} f_n q^n$,*

$$\widehat{f} \in \mathcal{C}_{\mathbb{M}} \quad \Longleftrightarrow \quad \widehat{f}^{(-1)} \in \mathcal{C}_{\mathbb{M}}.$$

Proof. Write $\hat{f}(q) = q(1 + \hat{h}(q))$ and its proposed inverse as $q = zW(z)$, $W(0) = 1$. The inverse equation is

$$W = 1 - W\hat{h}(zW). \quad (9.1)$$

The coefficient of each positive power of z is a polynomial in W . On a fixed disk about 1 it is bounded by a geometric factor times the corresponding coefficient of $\hat{h}$. Division by q preserves the weight class, so the formal-majorant step of Theorem 3.1 proves $W \in \mathcal{C}_{\mathbb{M}}$. Multiplication by z proves one implication; applying it again to the inverse proves the other. $\square$

Theorem 9.2 (Strong inverse realization). *There is a canonical sectorial inverse Q on a left sectorial germ, with $Q(z) = z + O(z^2)$. For every admissible weight,*

$$\hat{U} \in \mathcal{C}_{\mathbb{M}} \iff \hat{Q} \in \mathcal{C}_{\mathbb{M}} \iff Q \text{ has strong } \mathbb{M}\text{-asymptotics}. \quad (9.2)$$

The Gevrey, summability, and optimal logarithmic-scale conclusions all hold for the inverse. When directional sums are asserted, they are actual compositional inverses.

Proof. Choose nested proper left sectors. On the larger one $U(q) - q = O(q^2)$; Cauchy's estimate on disks of radius proportional to $|q|$ gives $U'(q) - 1 = O(q)$ on the smaller one. For z in a still smaller sector, solve $q = z - (U(q) - q)$ in a disk centered at z of radius $K|z|^2$. For fixed sufficiently large K and small z , the disk stays in the preceding sector, the map preserves it, and its derivative has modulus less than $1/2$. This constructs a holomorphic $Q = z + O(z^2)$ satisfying $U(Q(z)) = z$. Local inverse uniqueness gives the other composition identity on compatible domains and the compatibility of sectorial restrictions.

Suppose now $\hat{U} \in \mathcal{C}_{\mathbb{M}}$. Theorem 5.2 gives a strong $\mathbb{M}$ -expansion for $V = U/q$. Choose a sufficiently small fixed disk $D(1, \delta)$ so that zv stays in the larger sector for z in the chosen smaller sector and v in that disk. For $Q = zW$, the equation is

$$W = 1 - W(V(zW) - 1). \quad (9.3)$$

If $V(q) \sim \sum v_n q^n$, the right side has coefficient polynomials $-v_n W^{n+1}$ for $n \geq 1$. These satisfy the required weighted estimates on the fixed disk. Its remainder is bounded by $CA^N M_N |zW|^N |W|$, hence by the same weight with a geometric enlargement. Theorem 3.1 yields strong asymptotics for W , and then for Q .

Strong asymptotics imply $\hat{Q} \in \mathcal{C}_{\mathbb{M}}$, and formal weighted reversion returns $\hat{U} \in \mathcal{C}_{\mathbb{M}}$. A smaller weight for Q would therefore give a smaller weight for U , contrary to Theorem 6.4. The summability statement follows by the same sector-opening argument and uniqueness of the corresponding sum. $\square$

Remark 9.3 (Cancellation is not ignored). We have proved weight-class optimality for the inverse, not a pointwise asymptotic formula for its signed coefficients. Equation (6.5) is proved for the positive coefficients of $\hat{U}$ and is not silently transferred to $\hat{Q}$. Equality of their sharp numerical Gevrey types is likewise not asserted.

10 Finite-action evaluation and executable checks

10.1 A uniform analytic certificate

Define

$$F_L(q, v) = \sum_{j=1}^L q^{j-1} e^{\lambda_j qv}, \quad V_L = F_L(q, V_L). \quad (10.1)$$

The same disk and derivative bounds hold. One may choose $\delta = \eta/4$, $c = 3\eta/4$, $B = 1 + \delta$, and a radius such that

$$\lambda_1 Br + \frac{r}{1-r} \leq \delta/2, \quad \kappa(r) := \lambda_1 r + \frac{r}{ec(1-r)} < 1. \quad (10.2)$$

Both finite and infinite maps preserve the chosen disk and have Lipschitz constant at most $\kappa(r)$.

Proposition 10.1 (Action and iteration error bounds). *Under (10.2), with $|q| = r$,*

$$|U(q) - qV_L(q)| \leq \frac{r^{L+1}}{(1 - \kappa(r))(1 - r)}. \quad (10.3)$$

If $v_0 = 1$, $v_{m+1} = F_L(q, v_m)$, then

$$|U(q) - qv_m| \leq \frac{r\kappa(r)^m}{1 - \kappa(r)}|v_1 - v_0| + \frac{r^{L+1}}{(1 - \kappa(r))(1 - r)}. \quad (10.4)$$

The action-tail bound and κ are independent of the slopes λ_j for $j \geq 2$. In the iteration bound, replacing the computed increment $|v_1 - v_0|$ by $\lambda_1 Br + r/(1 - r)$ makes the entire right side independent of those slopes.

Proof. The kernel tail is at most $r^L/(1 - r)$. Subtract the fixed-point equations, use contraction, and multiply by r to prove (10.3). Summing the geometric bound on successive iteration increments gives $|V_L - v_m| \leq \kappa^m|v_1 - v_0|/(1 - \kappa)$; add the estimates. $\square$

Finite-action truncation and Taylor truncation are different. The former converges geometrically at fixed q even when the Taylor series diverges; the latter has the optimally chosen error bounds proved above. For end-to-end certified numerics the finite exponential evaluations and the right side of (10.4) must also be enclosed with rounding errors. The supplied high-precision diagnostics do not use interval arithmetic. Their reported error bounds are floating-point evaluations of the proved analytic certificate.

10.2 An integer recurrence

For integer slopes let $b_n = n!u_n$ and $E_{j,n} = n![q^n]e^{\lambda_j \widehat{U}(q)}$, with $E_{j,0} = 1$. Formal differentiation gives

$$b_n = \sum_{j=1}^n \frac{n!}{(n-j)!} E_{j,n-j}, \quad (10.5)$$

$$E_{j,n} = \lambda_j \sum_{m=1}^n \binom{n-1}{m-1} b_m E_{j,n-m}. \quad (10.6)$$

At stage n the first line uses previously computed exponential coefficients; the second is then evaluated for every needed j . All arithmetic is integral. Independent partition enumeration checks (5.1) at small orders.

10.3 Recorded verification

The bundled program was executed through degree 120 for each of $\lambda_j = 0, j, j^2, j^3$. Its exact checks were:

Check	Number passed
Independent partition-formula comparisons	48
Positive-subfamily inequalities $a_n(1) \leq u_{n+1}$	480
Independent zero- and linear-slope benchmarks	240
Total exact comparisons	768

All passed. The linear benchmark uses the separate identity $U = q(1+U)e^U$ and its one-variable Lagrange coefficient formula. Full exact integer-normalized coefficients are included in CSV files. For the quadratic model,

$$\begin{aligned}\widehat{U}(q) = & q + 2q^2 + \frac{15}{2}q^3 + \frac{107}{3}q^4 + \frac{1555}{8}q^5 + \frac{17362}{15}q^6 \\ & + \frac{5288311}{720}q^7 + \frac{15414767}{315}q^8 + \cdots ;\end{aligned}$$

for the cubic model the first terms are

$$q + 2q^2 + \frac{23}{2}q^3 + \frac{269}{3}q^4 + \frac{21433}{24}q^5 + \frac{53144}{5}q^6 + \cdots .$$

At 100 decimal digits the program also checked the kernel remainder inequality in 48 cases and compared six finite-action solves with longer, more accurate solves. The reported approximations use $L = 18$, $m = 24$. A sample is

p	q	Approximate $U(q)$	Analytic bound
2	-0.01	-0.0098071616822383391168465501	$1.031 \cdot 10^{-38}$
2	-0.005	-0.0049509157982251340457345151	$1.937 \cdot 10^{-44}$
3	-0.01	-0.0098106832787087467452630685	$1.031 \cdot 10^{-38}$
3	-0.005	-0.0049513840936461045663286861	$1.937 \cdot 10^{-44}$

The bound concerns the internally retained high-precision approximation, not the shortened printed decimal. CSV files record additional digits, complex tests, and observed comparison errors. Numerical agreement does not prove an asymptotic or summability theorem.

11 Examples and boundary cases

For $\lambda_j = j^{3/2}$ the optimal weight has $s = 1/2$, $\tau = -3/2$. The series is 2-summable in direction π , with sum equal to the literal left-sector fixed point. Optimized Taylor truncation gives

$$O(\exp[-c|q|^{-2}(\log(1/|q|))^3]).$$

The logarithmically refined sum records the sharper natural weight.

Fix $k > 1$. For slopes $(j \log(j+1))^{1+1/k}$, negative-direction k -summability holds. Multiplication by $(\log(j+1))^\varepsilon$, $\varepsilon > 0$, destroys it by the necessary slope criterion. The sectorial solution still exists, and a logarithmically refined summation method or slightly smaller ordinary summation order remains available.

For $\lambda_j = j^2$, strong Gevrey-1 asymptotics and (7.5) are proved, but ordinary negative-direction Borel summability is not decided here. For $\lambda_j = j^3$, the natural weight is $M_{2,-3}$, giving

$$O(\exp[-c|q|^{-1/2}(\log(1/|q|))^{3/2}]).$$

The directly available sector opening does not suffice to deduce summability at the critical order. No failure of summability is inferred merely from that limitation of the argument.

For $\lambda_j = e^{\sqrt{j}}$, the slope condition fails for every finite $s > 0$. No proper left sector therefore has a strong finite-order Gevrey expansion of the canonical solution. The sectorial existence theorem and finite-action certificate nevertheless remain valid. Other weight classes may apply, but their hypotheses must be checked.

Slopes	Strong remainder information	Summability proved here
$O(j)$	Convergent Taylor series	Analytic germ
$j \log(j+1)$	Every positive Gevrey class	Every $k > 1$, direction π
$j^p, 1 < p < 2$	$M_{p-1,-p}$, optimal in scale	Critical $k = 1/(p-1)$ and refined sum
j^2	$M_{1,-2}$ and (7.5)	Critical case left open
j^3	$M_{2,-3}$, optimal in scale	No critical-order conclusion
$e^{\sqrt{j}}$	No finite Gevrey order	No finite-order claim

After the first row all formal series are divergent. Every row still has the canonical left-sector solution.

12 Proof status and further research

12.1 Statement-level proof ledger

The sectorial construction, kernel remainder estimate, weighted implicit theorem, positive coefficient comparison, coefficient-to-remainder equivalence, Gevrey criterion, logarithmic optimality, action certificate, and strong inverse theorem have self-contained proofs here. The summability conclusions use the explicitly cited classical sectorial characterization and generalized reconstruction theorem. Formal Lagrange inversion, Cauchy estimates, and contraction are used in their standard forms.

No statement is claimed Lean-verified. The repository’s formal inventory does not certify this article, and numerical testing does not upgrade its proofs into machine-checked mathematics. The sharper exact numerical type formulas and saddle equivalents in neighboring drafts are not needed and have not been audited here. The advance is specific: formal coefficient classes now control the canonical sectorial remainders, and the subquadratic-opening regime gives an exact directional summability classification. This is not a solution to analytic realization in arbitrary transseries fields.

12.2 A formalization decomposition

The finite layer consists of the integer recurrence, partition formula, marked positive subfamily, and log-convex weight inequalities. It needs ordinary formal power series and finite partitions, not a new transseries datatype. The analytic layer consists of the half-plane exponential Taylor bound, compact normal convergence, fixed-disk contraction, and the shrinking-disk weighted implicit theorem. A further module can package coefficient-to-remainder equivalence. A summability layer should import an explicit sectorial characterization rather than imply that existing power-series APIs already prove Borel reconstruction.

12.3 Further questions with concrete targets

1. Critical negative-ray Borel summability. Decide ordinary negative-direction Borel summability for quadratic feedback. The missing step is uniform control on an appropriate boundary domain, not another coefficient-growth estimate. Inverse coordinates, where the

literal kernel is analytic for $\operatorname{Re} U < 0$, may help construct a Nevanlinna–Sokal domain after returning to q .

Editorial note (ProveIt, 2026-09-29). The inversion-first method proposed here is the method of `Negative_Ray_Summation_Exponential_Feedback/` [8], which uses it to prove fine summability in direction π for $\lambda_j = O((j \log j)^2)$. Ordinary (angular) summability for $\lambda_j = j^2$ fails by `Natural_Boundaries_Quadratic_Exponential_Feedback/` [9].

2. Sharp numerical remainder types. Compare the infimal geometric constant in strong remainders with the coefficient type. A no-loss argument could sharpen the exponential constant in (7.5) and determine whether precise type is invariant under the inverse. Our proof establishes class equality only.

Editorial note (ProveIt, 2026-09-29). For coefficient types this is answered by `Sharp_Weighted_Type_Formal_Reversion/` [10]: the exact Gevrey type is invariant under the inverse (its Theorem `thm:inversefeedback`). The sharp constant in the remainder bounds remains open.

3. Curved boundary layers. Since $U = q + u_2 q^2 + \dots$ with $u_2 > 0$, the boundary $\operatorname{Re} U = 0$ need not be $\operatorname{Re} q = 0$. Analyze the layer $\operatorname{Re} q = O(|q|^2)$ and the maximal domain of the literal canonical solution. A fixed v -disk does not provide uniformity there; a curved or shrinking dependent-variable domain may be required.

4. Summation and acceleration for $p \geq 2$. The natural opening index is $p - 1$, while the directly constructed germ has opening 1 in units of π . Investigate continuation, acceleration, and singular barriers. Divergence of the literal positive-real kernel does not settle analytically continued or generalized summation questions.

5. Signed and complex feedback. For $U = \sum c_j q^j e^{\lambda_j U}$, find hypotheses replacing the positive subfamily estimate. Absolute convergence can preserve the analytic disk argument, but coefficient cancellations may conceal large kernel terms. Identify the exact obstruction to coefficient-to-remainder equivalence.

6. Vector systems. For $U_i = \sum_j q^{a_{ij}} c_{ij} \exp(\langle \lambda_{ij}, U \rangle)$, seek a common dissipative cone, normalize each leading component, and prove a multitype counterpart of the bridge. The positivity argument would require a marked-tree comparison that respects the coupled components.

7. Faster weights. The hypothesis $\mu_n^n \leq D^n M_n$ excludes weights such as $M_n = a^{n^2}$, $a > 1$. Determine whether a modified shrinking-domain argument yields a bridge in these regimes or whether the feedback model provides counterexamples. This localizes a potential boundary of control by formal coefficient growth.

8. Fully verified evaluation. Combine exact coefficients, interval exponential evaluation, the action certificate, and effective weighted constants into an algorithm choosing between action truncation and Taylor truncation. Its output should record the method and provide a machine-checkable numerical enclosure.

13 Conclusion

The original coordinates really do lack an analytic dependent-variable neighborhood when feedback slopes grow too rapidly. That failure does not prevent uniform sectorial asymptotics. Factoring out the leading variable exposes a fixed disk on which the exponentials are bounded; positivity identifies its remainders with a controlled subfamily of the formal coefficients. This proves an exact bridge from formal regularity to canonical strong sectorial regularity, optimal factorial–logarithmic classes, and a complete negative-direction k -summability criterion for $k > 1$. The actual inverse shares the structure. Remaining problems are now localized to sharp constants, critical aperture, angular continuation, cancellation, and faster weights.

A Reproduction and file inventory

The source is self-contained and loads no repository notation files. A standard TeX installation with the preamble packages suffices:

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The exact checks require only Python’s standard library; optional numerical checks use mpmath:

```
python code/verify.py --order 120 --exact-only
python -m pip install mpmath
python code/verify.py --order 120
```

The full run regenerates `data/verification.json`, four exact coefficient CSVs, and the sectorial diagnostics CSV. The CSV column `n_factorial_times_u_n` contains $n!u_n$, not u_n . Numerator and denominator columns give the ordinary coefficients exactly. The package also contains a README, source/provenance notes, proof-status summary, recorded output, and a Makefile. No repository changes were made and no external font files are distributed.

References

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- [8] [Editorial addition, ProveIt, 2026-09-29.] *Negative-Ray Summation of Countable Exponential-Feedback Transseries*, research article prepared for V. Reshetnikov, 29 September 2026. In [1], directory `Negative_Ray_Summation_Exponential_Feedback/`; present at the snapshot used by this article.
- [9] [Editorial addition, ProveIt, 2026-09-29.] *Natural Boundaries of Quadratic Exponential Feedback*, research draft prepared for V. Reshetnikov, 29 September 2026. Directory `Natural_Boundaries_Quadratic_Exponential_Feedback/` of the same series; filed after this article.
- [10] [Editorial addition, ProveIt, 2026-09-29.] *Sharp Weighted Type Under Formal Reversion*, research article prepared for V. Reshetnikov, 29 September 2026. Directory `Sharp_Weighted_Type_Formal_Reversion/` of the same series; filed with this article.